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**ITU-T** **V.cvm**

TELECOMMUNICATION STANDARDIZATION SECTOR OF ITU (October 2026)

SERIES V: DATA COMMUNICATION OVER THE TELEPHONE NETWORK

Interworking with other networks

# Data transmission over the speech paths of mobile telephony networks

Draft new Recommendation ITU-T V.cvm

Proposal

**Status of this document.** This document is a *proposal* for a new Recommendation, written in the form and the conventions of an ITU-T Recommendation so that it may be contributed to the ITU-T study group responsible for the V series. It has not been approved by ITU-T, its designation V.cvm is a working name, and nothing in it binds ITU-T. The capability bits in the V.8 and V.8bis signals that it needs (Annex A) are allocations that only ITU-T can make; until they are made, the non-standard facilities mechanism of those Recommendations is used, as Annex A says.

**Where its numbers come from.** The behaviour of the speech codecs towards modem signals, on which the design rests, was measured with the D-Modem project's modems and test tools through the AMR and AMR-WB codecs (the OpenCORE and VisualOn implementations of 3GPP TS 26.090 and 26.190, checked bit-exact against the 3GPP reference code) and through the EVS codec (the 3GPP reference code of TS 26.442). Appendix I gives the measurements and how to repeat them; Appendix II those of EVS.

## Summary

Modern mobile telephony carries speech as packets of a low bit rate speech codec: AMR in GSM and narrowband VoLTE, AMR-WB (ITU-T G.722.2) in wideband VoLTE, EVS in VoLTE and 5G voice (VoNR). These codecs model speech, not waveforms, and the voice-band modems of the V series fail through them: of the modulations of Recommendations ITU-T V.21 to V.34, only V.21 and V.22 come through the AMR codecs at all, and at 300 and 1200 bit/s. A circuit-switched data bearer is no longer offered. Where the only path between two parties is a telephone call – the general case between networks, operators and countries – data must go through the codec.

Recommendation ITU-T V.cvm specifies a data circuit-terminating equipment (DCE) for that path. It reuses the V series wherever the codec leaves it usable and replaces only what the codec breaks:

- the call starts as a V.8 session (ANSam, CI, CM, JM in V.21), with an indication of this Recommendation's capabilities, and falls back to V.22 or V.21 towards a DCE that does not have them;
- the line is probed, in the manner of V.34's line probing, for the codec's bandwidth and its noise against a multitone signal, and a mode is chosen from the measurements;
- the signal is a comb of tones with differential phase modulation, a symbol every codec frame or two, in the band the codec carries; the tones are loaded by the probing, and the mode is renegotiated as the codec's mode changes, in the manner of V.34's rate renegotiation;
- a Reed-Solomon code with interleaving (the code of ITU-T G.975) corrects the codec's errors and the erasures of lost frames, and V.42's LAPM, V.42bis and V.44 run above it unchanged;
- the DTE interface is V.250's, with the additions this Recommendation needs; facsimile is carried as T.38 packets.

The measured raw rate of the signal set is 1600 bit/s through AMR at 12.2 kbit/s, 800 through EVS at 13.2 kbit/s, 2200 to 2800 through AMR-WB and EVS at 24.4 kbit/s wideband, 4950 through EVS at 48 kbit/s and 10 350 through EVS at 128 kbit/s super wideband; the bound for any signal set is the codec's own bit rate. Through AMR at 7.4 kbit/s and below, only the 300 bit/s of V.21 passes.

## Keywords

AMR, AMR-WB, data over voice, EVS, modem, speech codec, V.22, V.8, VoLTE, VoNR.

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## 1 Scope

This Recommendation specifies a DCE for data transmission over the speech path of a telephone call carried by a mobile telephony network, that is, through one or more low bit rate speech codecs: the adaptive multi-rate codec (AMR, [3GPP TS 26.090]), the adaptive multi-rate wideband codec (AMR-WB, [ITU-T G.722.2], [3GPP TS 26.190]) and the enhanced voice services codec (EVS, [3GPP TS 26.445]), as the IP multimedia subsystem carries them in VoLTE and VoNR calls ([3GPP TS 26.114]), and through the circuit-switched speech bearers of earlier networks. It specifies:

- the reference configuration and the interface at which the DCE's signal enters and leaves the speech path (clause 7);
- the start-up procedures, built on [ITU-T V.8] (clause 8);
- the probing of the path and the selection of a mode of operation (clause 9);
- the signal set: its modulation modes, symbol timing and tone loading (clause 10);
- error control, framing and the use of [ITU-T V.42], [ITU-T V.42bis] and [ITU-T V.44] above it (clause 11);
- the DTE interface, as additions to [ITU-T V.250] (clause 12);
- interworking with the V series DCEs of the PSTN and with facsimile (clause 13, Annex D).

It does not specify the speech codecs, the mobile network, the terminal's acoustic processing or the IMS signalling, and it does not change the speech path in any way: it is a signal the path carries; what the terminal and the network should do for the DCE is in Appendix III (informative). Data transmission over the packet data bearers of the same networks is out of scope: this Recommendation is for the case where the telephone call is the path available.

## 2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T G.711] Recommendation ITU-T G.711 (1988), *Pulse code modulation (PCM) of voice frequencies*.

[ITU-T G.722.2] Recommendation ITU-T G.722.2 (2003), *Wideband coding of speech at around 16 kbit/s using Adaptive Multi-Rate Wideband (AMR-WB)*.

[ITU-T G.975] Recommendation ITU-T G.975 (2000), *Forward error correction for submarine systems*.

[ITU-T T.30] Recommendation ITU-T T.30 (2005), *Procedures for document facsimile transmission in the general switched telephone network*.

[ITU-T T.38] Recommendation ITU-T T.38 (2010), *Procedures for real-time Group 3 facsimile communication over IP networks*.

[ITU-T V.8] Recommendation ITU-T V.8 (2000), *Procedures for starting sessions of data transmission over the public switched telephone network*.

[ITU-T V.8bis] Recommendation ITU-T V.8bis (2000), *Procedures for the identification and selection of common modes of operation between data circuit-terminating equipments (DCEs) and between data terminal equipments (DTEs) over the public switched telephone network and on leased point-to-point telephone-type circuits*.

[ITU-T V.21] Recommendation ITU-T V.21 (1988), *300 bits per second duplex modem standardized for use in the general switched telephone network*.

[ITU-T V.22] Recommendation ITU-T V.22 (1988), *1200 bits per second duplex modem stan-*

*dardized for use in the general switched telephone network and on point-to-point 2-wire leased telephone-type circuits*.

[ITU-T V.34] Recommendation ITU-T V.34 (1998), *A modem operating at data signalling rates of up to 33 600 bit/s for use on the general switched telephone network and on leased point-to-point 2-wire telephone-type circuits*.

[ITU-T V.42] Recommendation ITU-T V.42 (2002), *Error-correcting procedures for DCEs using asynchronous-to-synchronous conversion*.

[ITU-T V.42bis] Recommendation ITU-T V.42bis (1990), *Data compression procedures for data circuit-terminating equipment (DCE) using error correction procedures*.

[ITU-T V.44] Recommendation ITU-T V.44 (2000), *Data compression procedures*.

[ITU-T V.250] Recommendation ITU-T V.250 (2003), *Serial asynchronous automatic dialling and control*.

[ITU-T T.35] Recommendation ITU-T T.35 (2000), *Procedure for the allocation of ITU-T defined codes for non-standard facilities*.

[3GPP TS 26.090] 3GPP TS 26.090, *Mandatory Speech Codec speech processing functions; Adaptive Multi-Rate (AMR) speech codec; Transcoding functions*.

[3GPP TS 26.190] 3GPP TS 26.190, *Speech codec speech processing functions; Adaptive Multi-Rate – Wideband (AMR-WB) speech codec; Transcoding functions*.

[3GPP TS 26.445] 3GPP TS 26.445, *Codec for Enhanced Voice Services (EVS); Detailed algorithmic description*.

[3GPP TS 26.114] 3GPP TS 26.114, *IP Multimedia Subsystem (IMS); Multimedia telephony; Media handling and interaction*.

## 3 Definitions

### 3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

**answering DCE, calling DCE** [ITU-T V.8].

**call function, modulation mode** [ITU-T V.8]: the categories of the CM and JM signals.

**frame erasure** [3GPP TS 26.090]: a codec frame that the decoder does not receive, or receives marked as bad, and conceals.

**LAPM** [ITU-T V.42]: the link access procedure for modems.

**line probing** [ITU-T V.34]: the transmission of a known multitone signal so that the receiver may measure the channel.

**rate renegotiation** [ITU-T V.34]: a change of the data signalling rate during data transmission, without retraining.

### 3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

**speech path** the path from the calling DCE's signal interface to the answering DCE's, through the terminals' speech processing, one or more speech codecs and the networks between them. It is the channel of this Recommendation.

**signal interface** the point at which the DCE's transmitted signal enters the speech path and the received signal leaves it, as linear PCM samples (reference point P, clause 7).

**codec frame** the block of samples a speech codec encodes together: 20 ms for the codecs of this Recommendation.

**comb** the set of tones of the signal set, at a common spacing.

**tone symbol** the information carried by one tone in one modulation symbol: a phase difference from the tone's previous symbol.

**mode** a signal set, a comb, a symbol period, a modulation order and a code rate, agreed between the DCEs by the probing procedure.

**loading** the set of tones in use and the modulation order of each.

## 4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

| | |
|---|---|
| ACELP | Algebraic code excited linear prediction |
| AMR | Adaptive multi-rate (speech codec) |
| AMR-WB | Adaptive multi-rate wideband (speech codec) |
| ANSam | Modified answer tone ([ITU-T V.8]) |
| CI | Call indicator signal ([ITU-T V.8]) |
| CM, JM | Call menu, joint menu signals ([ITU-T V.8]) |
| DCE | Data circuit-terminating equipment |
| DPSK | Differential phase shift keying |
| DTE | Data terminal equipment |
| DTX | Discontinuous transmission |
| EVS | Enhanced voice services (speech codec) |
| FEC | Forward error correction |
| IMS | IP multimedia subsystem |
| JBM | Jitter buffer management |
| LAPM | Link access procedure for modems |
| LPC | Linear predictive coding |
| MDCT | Modified discrete cosine transform |
| PCM | Pulse code modulation |
| PLC | Packet loss concealment |
| PSTN | Public switched telephone network |
| RS | Reed-Solomon (code) |
| SDP | Session description protocol |
| SER | Symbol error ratio |
| UE | User equipment |
| VAD | Voice activity detection |
| VoLTE, VoNR | Voice over LTE, voice over new radio (5G) |

## 5 Conventions

In this Recommendation, “shall” indicates a requirement, “should” a recommendation and “may” a permission, as in ITU-T practice. Data signalling rates are in bit/s; the rate before the Reed-Solomon code is the *raw* rate and the rate after it the *coded* rate. Levels are relative to the full scale of the 16 bit linear PCM samples at the signal interface (dBFS); −12 dBFS is the level of the test signals. Frequencies of tones are in Hz. Octets are shown most significant bit first; bit 1 is the least significant. Figures and tables give the values of this Recommendation; where a value was measured, the measurement is in Appendix I.

## 6 General description

### 6.1 The speech path as a channel

A speech codec of the ACELP family (AMR, AMR-WB, the ACELP modes of EVS) describes each 20 ms frame of its input by a linear predictive spectral envelope, a long-term (pitch) predictor with a lag and a gain per 5 ms subframe, and a sparse algebraic excitation of a few pulses per subframe. The decoder synthesizes a signal from these parameters. The result matches the input’s waveform only as far as the

parameters can describe it and as far as the perceptually weighted error criterion of the encoder cares to. For a modem's signal the consequences, measured in Appendix I, are:

a) the signal-to-noise ratio of the output against the input is low, in the order of 10 to 15 dB for signals with the spectrum of speech, and it does not improve with the input level: the noise is the codec's, proportional to the signal;
b) a signal of many tones with a smooth spectral envelope is carried better than a single tone, whose line spectrum the predictive model cannot place exactly; single-carrier modulations therefore fare worse than their bandwidth suggests, and multitone signals better;
c) the phase of each tone is carried from one frame to the next well enough for differential phase modulation of four states per tone and symbol, when the symbol lasts a codec frame or two, and not for eight states or for amplitude modulation;
d) the band is the codec's: 200 to 3400 Hz for AMR, 50 to 6400 Hz for AMR-WB (its 6.4 to 7 kHz is a bandwidth extension that carries no waveform), and for EVS by its bandwidth mode and bit rate: at 24.4 kbit/s to about 8 kHz and at 48 kbit/s and above to 12 or 14 kHz (Appendix II);
e) a lost packet erases a frame; the decoder conceals it with an extrapolation, which to the modem is 20 ms of noise; the jitter buffer of the receiving terminal may also repeat or drop a frame, or stretch the signal, to adapt its delay;
f) the codec's mode (its bit rate) is chosen by the network and changes during the call; the noise changes with it;
g) the one-way delay is 100 to 300 ms, and differs between the two directions.

Nothing in the path is a linear channel; the effects of a) to c) are those of a channel with a noise floor some 10 to 15 dB below the signal, and the design of this Recommendation is a modem for such a channel, with the measures of e) to g) around it.

NOTE – The throughput of any signal set through a codec is bounded by the codec's bit rate: 12.2 kbit/s for AMR at its highest mode, 23.85 for AMR-WB, 13.2 to 128 kbit/s for EVS. The codec's parameters describe the signal, and no more information passes than they carry. Practical signal sets reach a fraction of the bound; the one of this Recommendation reaches in the order of 10 to 15 per cent of it through AMR and EVS.

### 6.2 What this Recommendation reuses

Table 1 lists the parts of the V series and of other Recommendations that this Recommendation uses as they are, and what it replaces.

### 6.3 Overview of a call

Figure 1 shows the phases of a call. The answering DCE sends ANSam; the calling DCE sends CI then CM with this Recommendation's capabilities; the answering DCE answers with JM. If both have the capability, the DCEs exchange probing signals and INFO messages (phase 2), each chooses a loading for the direction it receives and tells the other (phase 3, MP), and data transmission begins (phase 4). During data transmission either DCE may ask for a renegotiation of its loading or a retrain. If either DCE lacks the capability, the call continues as a [ITU-T V.8] session to V.22 or V.21 as the common modulation modes allow.

## 7 Reference configuration and interfaces

### 7.1 Reference configuration

Figure 2 gives the reference configuration. Reference point P is the signal interface: the DCE gives its transmitted signal to the terminal as linear PCM samples and takes the received signal from it in the same form. The terminal is whatever carries the speech path: a UE, an IMS client, a gateway between the IMS and another network, or a circuit-switched mobile terminal. Everything from one reference point P to the other is the speech path.

**Table 1** – The V series and other Recommendations in this Recommendation

| Function | Reused | New in this Recommendation |
|---|---|---|
| Start-up | [ITU-T V.8]: ANSam, CI, CM, JM, CJ, in V.21 (300 bit/s FSK, which the codecs carry) | capability indication (Annex A) |
| Fallback | [ITU-T V.22] (1200 bit/s, through AMR 12.2) and [ITU-T V.21] (300 bit/s, through every mode), as they are | – |
| Probing | the principle of [ITU-T V.34] clause 10: a multitone probing signal, INFO exchanges, MP parameter exchange, rate renegotiation and retrain | the probing signals and the parameters exchanged (clause 9) |
| Modulation | the differential phase modulation of [ITU-T V.22] per tone; the multitone structure of ADSL ([ITU-T G.992.1]) with a cyclic prefix | the comb, symbol timing and loading (clause 10) |
| Inner code | the Reed-Solomon code of [ITU-T G.975] over GF($2^8$), shortened | its interleaving over symbols and frames (Annex C) |
| Link layer | [ITU-T V.42] LAPM with its detection phase, [ITU-T V.42bis] and [ITU-T V.44] compression | – |
| DTE interface | [ITU-T V.250] | the +MS values, S-registers and result codes of clause 12 |
| Facsimile | [ITU-T T.38] internet facsimile packets, [ITU-T T.30] | their carriage in the data link (Annex D) |
| Codec | [ITU-T G.722.2] (AMR-WB) as a channel model | – |

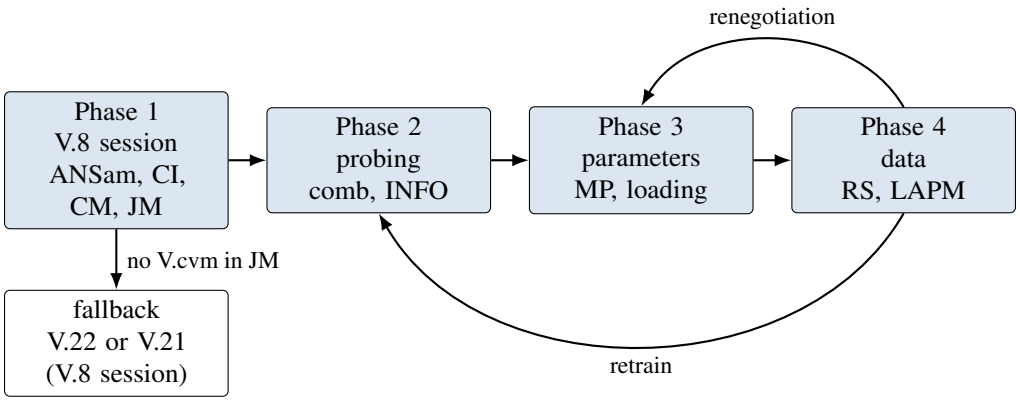



**Figure 1** – The phases of a call

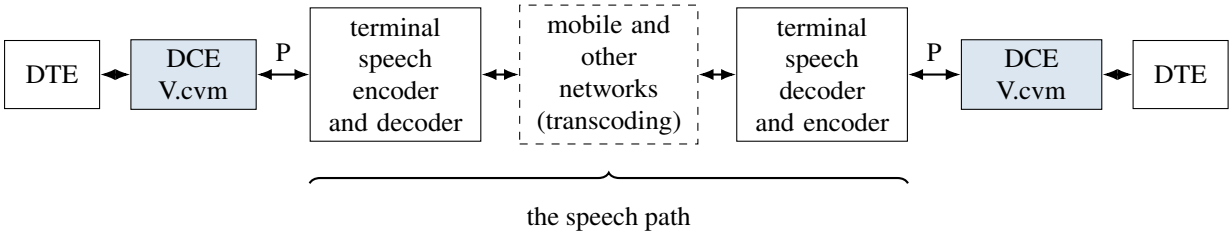



**Figure 2** – Reference configuration

### 7.2 The signal interface (reference point P)

At reference point P:

a) samples shall be 16 bit linear PCM, two's complement, at 8000 Hz (narrowband operation) or 16000 Hz (wideband operation). A DCE shall support 8000 Hz and may support 16000 Hz. The rate is the terminal's; the DCE learns the band the path carries by probing, not from the rate.
b) the DCE's transmitted signal shall not exceed −9 dBFS in any 20 ms frame and should be at $-12 \pm 2$ dBFS in the data phases, so that no terminal gain stage limits it and the codec's gain quantizers are in their range. The received level is whatever the path delivers; the DCE shall accept −30 to −3 dBFS.
c) the terminal should pass the samples to and from the speech codec without acoustic processing (echo cancellation, noise suppression, automatic gain control, voice activity detection with comfort noise) between P and the codec. Where the terminal cannot disable them, the DCE shall still operate, with the modes the probing finds usable.
d) the sample clocks of the two terminals are independent. The DCE shall tolerate a difference of ±200 ppm between its transmit clock and the received signal's, and the steps of clause 10.3.

NOTE – A DCE in a UE must reach the speech codec's PCM interface. The acoustic path (loudspeaker to microphone) is not a signal interface of this Recommendation.

### 7.3 Interchange circuits

The DTE interface is the serial asynchronous interface of [ITU-T V.250] (clause 12), on any physical interface a V.250 DCE may have.

## 8 Start-up procedures

### 8.1 General

A DCE of this Recommendation shall perform the start-up procedures of [ITU-T V.8]: the answering DCE transmits ANSam; the calling DCE transmits CI and, on detecting ANSam, CM; the answering DCE answers with JM; the calling DCE acknowledges with CJ. The V.21 modulation of CI, CM, JM and CJ passes through every mode of the codecs (Appendix I, Table I.2), and ANSam's 2100 Hz tone, with its phase reversals, disables the network's echo cancellers (as it is meant to), which matters at a gateway to the PSTN.

### 8.2 Capability indication

The calling DCE shall indicate its support of this Recommendation in CM and the answering DCE in JM, as Annex A specifies, together with the modulation modes of its fallback (V.22 and V.21) and the protocol (LAPM). The indication carries:

- the version of this Recommendation (1);
- the signal interface rate: 8000, or 16000 Hz;
- the modes of clause 10 the DCE supports;
- whether the DCE can carry facsimile (Annex D).

### 8.3 Outcome

When both CM and JM carry the indication, both DCEs proceed to the probing of clause 9 on the completion of the V.8 procedures (CJ sent, or JM ended), within the time [ITU-T V.8] allows for the start of the selected modulation. Otherwise the DCEs proceed to the common modulation mode of highest rate among those indicated, V.22 then V.21, as [ITU-T V.8] and that Recommendation specify, and this Recommendation has nothing more to say about the call.

NOTE – A legacy V series DCE on the PSTN, reached through a gateway, indicates its own modulation modes (V.34, V.32bis, ...) and none of this Recommendation's. Of its modulation modes, V.22 is the fastest the codec carries (Table I.2): a DCE of this Recommendation should not indicate modulation modes above V.22 in CM when it knows the speech path to be a codec's, so that the legacy DCE selects V.22 at once rather than fail at V.34 and renegotiate.

### 8.4 Timing

The DCEs' timers for the V.8 procedures are [ITU-T V.8]'s. The probing of clause 9 shall begin within 200 ms of the end of the V.8 procedures and shall be complete, with the first data symbols sent, within 12 s (each direction's INFO0, L1, L2 and INFO1 take about 5 s at 300 bit/s for the messages); a DCE that has not received its peer's INFO0 within 6 s of the end of the V.8 procedures shall fall back to V.22 by restarting the V.8 procedures, once.

## 9 Probing and mode selection

### 9.1 Purpose

Probing measures the speech path as the signal set will see it:

a) the band carried, from which the signal interface rate of the far end and the codec's bandwidth mode follow;
b) the noise per tone against a multitone signal of the signal set's own form, from which the loading follows;
c) the round-trip delay, for the timers of the link layer.

Each direction is probed and loaded separately: the two directions of a mobile call go through different codecs, radio links and jitter buffers.

### 9.2 Phase 2: the probing signals

After the V.8 procedures each DCE transmits, in turn as Figure 3 shows:

1. **INFO0**: its signal interface rate, the modes it supports and the comb it will probe with, modulated as clause 9.3 specifies;
2. **L1**: the probing combs of Annex B: each comb the DCE supports for the band of its signal interface, in turn (200, 300 and 400 Hz spacing, Table 3), all tones loaded, each tone with a differential phase sequence known to both, 0.6 s a comb;
3. **L2**: the 200 Hz comb with the 22.5 ms symbol, for 0.5 s, so that the receiver may compare the two symbol periods;
4. silence for 100 ms, then **INFO1**: what it measured of its peer's L1 and L2 as the loading it asks for (clause 9.4), and its own round-trip estimate.

The calling DCE transmits first; the answering DCE begins its INFO0 on the end of the calling DCE's INFO1, as [ITU-T V.34] phase 2 orders INFO0c and INFO0a. The probing signal's level is −12 dBFS.

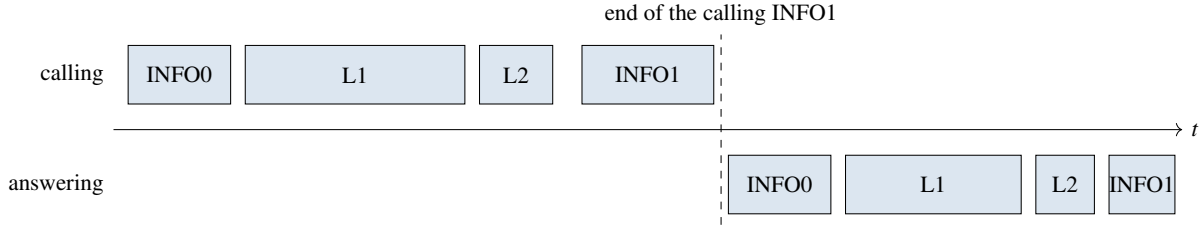



**Figure 3** – Phase 2: the probing signals (durations not to scale)

### 9.3 Modulation of the INFO and MP messages

INFO0, INFO1 and the MP messages of clause 9.5 shall be transmitted in the *robust mode* of clause 10.1: the FSK of [ITU-T V.21] at 300 bit/s, channel 1 from the calling DCE and channel 2 from the answering DCE as [ITU-T V.8] uses them for CM and JM, as HDLC frames (flags, FCS) of [ITU-T V.42] with the message's octets as the information field. Each message is sent twice; the receiver takes the first copy whose FCS checks. V.21's bit error ratio was measured below $10^{-2}$ through every mode of AMR but 4.75 kbit/s, of AMR-WB but 6.6, and of EVS but 9.6 wideband (Appendix I, Table I.3).

NOTE – V.34 modulates INFO and MP at 600 bit/s with differential phase modulation on two tones. Through AMR at 7.4 kbit/s and below no differential phase modulation measured kept a symbol error ratio below $10^{-1}$, while 300 bit/s FSK did (Table I.3); so this Recommendation sends its messages as V.8 sends CM and JM.

### 9.4 The loading

From its peer's L1 and L2 the receiving DCE shall estimate, for each tone of the comb and each of the two symbol periods, the root mean square phase error of the received tone symbols against the known sequence, and for each tone the mean received level. It shall then choose:

a) the band: the tones whose level is within 20 dB of the strongest tone's; tones outside are not loaded (this finds the codec's band);
b) the comb and the symbol period: those giving the greatest number of bits per second over the tones loaded by c); a comb of fewer tones, more widely spaced, wins where the codec's noise against each tone of a denser comb is too high (Appendix I);
c) the modulation order of each tone, from its phase error $\sigma$: DQPSK where $\sigma \leq 20°$, 8-DPSK where $\sigma \leq 9°$ and the mode set allows it, none where $\sigma > 20°$;
d) the code rate of the Reed-Solomon code (Annex C) from the measured tone symbol error ratio over the loaded tones: rate 7/8 below $10^{-3}$, 3/4 below $10^{-2}$, 1/2 otherwise.

The thresholds in c) give a tone symbol error ratio of about $10^{-2}$ at the threshold for Gaussian phase error, which the code of Annex C corrects to below $10^{-6}$ at rate 3/4. The loading is sent to the peer in INFO1 and MP as a bitmap of the comb with two bits a tone (none, DQPSK, 8-DPSK).

### 9.5 Phase 3: parameter exchange

Each DCE transmits an **MP** message with the loading it asks for in the direction it receives, the symbol period and the code rate, in the manner of [ITU-T V.34]'s MP sequence; it shall transmit MP until it has received its peer's MP with the acknowledgement bit set, then once more with its own acknowledgement bit set, and begin data transmission (phase 4) with the loading its peer asked for. Both directions enter phase 4 within 100 ms of each other.

### 9.6 Renegotiation and retrain

During phase 4, a DCE whose receiver measures a phase error or an error ratio over the thresholds of clause 9.4 on tones it loaded, or well under them on tones it did not load, should ask for a *renegotiation*: it sends the control sequence of clause 10.7 and the DCEs exchange MP messages in the robust mode with the new loading, then resume data; the code's interleaver is flushed and restarted. A DCE that loses the symbol timing or receives no decodable symbol for 2 s shall ask for a *retrain*: both DCEs return to phase 2. The codec's mode changes with the radio conditions, and the loading shall follow it; a renegotiation shall take no more than 500 ms and a DCE shall not refuse one.

# 10 Modulation

## 10.1 Signal set

The signal set has three modes, Table 2. A DCE shall support the robust mode and the narrowband mode; it may support the wideband mode (which needs the signal interface at 16000 Hz).

**Table 2** – The modes of the signal set

| | **Robust (R)** | **Narrowband (N)** | **Wideband (W)** |
|---|---|---|---|
| Signal | the FSK of [ITU-T V.21]: channel 1 (980/1180 Hz) from the calling DCE, channel 2 (1650/1850 Hz) from the answering DCE, 300 bit/s | a comb of tones at 200, 300 or 400 Hz spacing in the band 300 to 3400 Hz (16, 11 or 8 tones), DPSK | a comb of tones at 200, 300 or 400 Hz spacing in the band 300 to 6900 Hz (33, 23 or 17 tones), DPSK |
| Symbol | 3.33 ms (one bit) | 20 ms (a codec frame): a useful part of whole cycles of the spacing and a cyclic prefix (clause 10.3); the 200 Hz comb also 22.5 ms | as N |
| Modulation | binary FSK | DQPSK, or 8-DPSK on tones with a low enough phase error | as N |
| Raw rate | 300 bit/s | 1600 bit/s (16 tones at 200 Hz, DQPSK, 20 ms), 1100 (11 at 300 Hz), 800 (8 at 400 Hz); half as much again with 8-DPSK | 3300 bit/s (33 tones at 200 Hz, DQPSK, 20 ms), 2300 (23 at 300 Hz), 1700 (17 at 400 Hz); half as much again with 8-DPSK |
| Use | INFO and MP messages, control sequences; data through codecs that carry nothing else | data | data |
| Passes (measured) | every AMR mode but 4.75, every AMR-WB mode but 6.6, every EVS mode but 9.6 wideband | AMR 12.2 (every comb), AMR 10.2 and two AMR 12.2 codecs in tandem (the 300 and 400 Hz combs), AMR-WB from 12.65, EVS 13.2 (N400) and 24.4 (every comb, 8-DPSK) | AMR-WB 23.85 (28 tones of W200), 15.85 (16 of W300), 12.65 (7 of W300); EVS 24.4 (W300), 48 and above (every comb, 8-DPSK); the super wideband combs through EVS at 48 kbit/s and above |

NOTE – The modes were chosen from measurements (Appendix I). Through AMR at 7.4 kbit/s and below, no differential phase modulation measured kept a symbol error ratio below $10^{-1}$, on one carrier or many; only 300 bit/s FSK did. Through AMR at 12.2 and AMR-WB at 23.85 kbit/s, a comb of 16 tones at 200 Hz spacing with DQPSK and a symbol of 20 to 22.5 ms kept its tone symbol error ratio at or below $10^{-2}$; combs of 32 tones at 100 Hz spacing did not (the codec's noise against each tone grows faster than the number of tones), and 8-DPSK held only on the better tones. Fewer tones, more widely spaced, keep working where more do not: through AMR at 10.2 kbit/s and through two AMR codecs in tandem the 11-tone comb at 300 Hz spacing kept $3 \cdot 10^{-3}$ while the 16-tone comb went to $3 \cdot 10^{-2}$, and through AMR-WB at 15.85 the 300 Hz comb loads 16 tones where the 200 Hz comb loads 10; so the signal set has three combs for each band and the probing chooses. A single carrier at 600 baud (V.22's) passes AMR 12.2 as well, at 1200 bit/s; the comb carries a third more at the same symbol error ratio and loads itself to the codec.

### 10.2 The comb

The comb's tones are at the frequencies

$$f_k = k \cdot \Delta f, \qquad k = k_{\min}, \ldots, k_{\max},$$

as Table 3 gives them. The narrowband combs N200, N300 and N400 are for the signal interface at

**Table 3** – The combs

| Code | Comb | $\Delta f$ (Hz) | $k$ | Tones | Band (Hz) | $T_u$ (ms) | $T_g$ (ms) |
|---|---|---|---|---|---|---|---|
| 0 | N200 | 200 | 2–17 | 16 | 400–3400 | 15 (or 20) | 5 (or 2.5) |
| 1 | N300 | 300 | 1–11 | 11 | 300–3300 | 10 | 10 |
| 2 | N400 | 400 | 1–8 | 8 | 400–3200 | 15 | 5 |
| 3 | W200 | 200 | 2–34 | 33 | 400–6800 | 15 | 5 |
| 4 | W300 | 300 | 1–23 | 23 | 300–6900 | 10 | 10 |
| 5 | W400 | 400 | 1–17 | 17 | 400–6800 | 15 | 5 |
| 6 | S200 | 200 | 2–69 | 68 | 400–13800 | 15 | 5 |
| 7 | S300 | 300 | 1–46 | 46 | 300–13800 | 10 | 10 |
| 8 | S400 | 400 | 1–35 | 35 | 400–14000 | 15 | 5 |
| 9–15 | – | reserved | | | | | |

8000 Hz, the wideband combs W200, W300 and W400 for 16000 Hz, and the super wideband combs S200, S300 and S400 for 32000 Hz (for the EVS codec's super wideband modes, Appendix II). The probing (clause 9) decides which tones are loaded; the transmitter sends nothing on a tone that is not. The transmitted signal in the useful part of symbol $n$ is

$$s_n(t) = A \sum_{k \in \mathscr{L}} \cos\left(2\pi f_k t + \varphi_{n,k}\right), \qquad 0 \le t < T_u,$$

where $\mathscr{L}$ is the set of loaded tones, $\varphi_{n,k}$ the tone's phase in this symbol and $A$ the amplitude that gives the level of clause 7.2 b) over the loaded tones. The useful period $T_u$ is a whole number of cycles of $\Delta f$, so that the tones are orthogonal over it.

### 10.3 Symbol timing and the cyclic prefix

A symbol consists of a cyclic prefix of $T_g$ followed by the useful part of $T_u$: the prefix is the last $T_g$ of the useful part, transmitted first, so that the signal is continuous across each symbol except for the phase steps at the symbol's start. The symbol periods are

- **20 ms**, the period of a codec frame, for every comb: $T_u$ and $T_g$ as Table 3 gives them (three cycles of the spacing for the 200 and 300 Hz combs, six for the 400 Hz combs), so that one symbol's phase steps fall in each frame and the prefix spans the codec's transient after them (a subframe of 5 ms, or two for the 300 Hz combs);
- **22.5 ms**, for the N200 comb only: $T_u = 20$ ms (four cycles), $T_g = 2.5$ ms, for codecs that gain from the longer integration (the probing compares the two).

The receiver shall integrate each tone over the useful part only. A receiver shall acquire the symbol timing from the cyclic prefix (the correlation of the signal with itself $T_u$ earlier peaks over the prefix) and shall re-acquire it within two symbols after a step of the received timing of up to $\pm 20$ ms, which a jitter buffer's adaptation causes (Appendix III). Differential modulation makes the symbol after such a step the only one lost.

### 10.4 The codec frame and the symbol

The symbols are not aligned to the codec's frames, and no alignment is signalled: the position of the symbols' phase steps within the frame was found to make no difference to the tone symbol error ratio

(Appendix I, Table I.6: the 20 ms symbol at a frame boundary and 10 ms off it), which the codec's 5 ms subframes and its long-term predictor explain: the step is coded in the subframe it falls in, wherever that is. The 20 ms symbol is chosen for its rate (12 per cent more than the 22.5 ms symbol) and for its match with the frame erasures: one erased frame is one symbol.

### 10.5 Differential phase modulation

Each loaded tone carries, per symbol, the phase difference $\varphi_{n,k} - \varphi_{n-1,k}$, one of four values for DQPSK or eight for 8-DPSK, mapped from the bits as Table 5 shows. The first symbol of phase 4, and the first after a renegotiation, carries no data: it is the phase reference. The receiver decides each tone symbol from the difference of the tone's phases in two consecutive symbols; no carrier recovery and no equalization are required, and a frequency offset between the DCEs' clocks of $\Delta f_c$ appears as a constant phase rotation of $2\pi\Delta f_c T$ per symbol, which the receiver should estimate from the decisions and remove.

### 10.6 Loading and raw rate

The raw rate is

$$R = \frac{1}{T_u + T_g} \sum_{k \in \mathscr{L}} b_k,$$

with $b_k = 2$ for a DQPSK tone and 3 for an 8-DPSK tone: 1600 bit/s for the 16 DQPSK tones of N200 at the 20 ms symbol, 1100 for N300, 800 for N400, 3300 for W200, and half as much again with 8-DPSK on all tones. The loading of clause 9.4 chooses $\mathscr{L}$ and the $b_k$; a loading with fewer than 4 tones shall not be used (the DCE renegotiates or retrains).

### 10.7 Control sequences during data

A DCE that wishes to renegotiate (clause 9.6) shall send, in place of data, the control sequence: eight consecutive symbols with the phase difference 180° on every loaded tone, then its MP in the robust mode. A receiver that decodes four such symbols in a row shall stop delivering data to the inner decoder, flush, answer with the same sequence if it has not sent one, and proceed to the MP exchange. The sequence cannot occur in scrambled data for more than two symbols with a probability above $10^{-9}$ on 8 tones.

### 10.8 Scrambling

The bits given to the mapping of clause 11.4 shall be scrambled with the generating polynomials of [ITU-T V.34]: $1 + x^{-18} + x^{-23}$ in the calling DCE's transmitter and $1 + x^{-5} + x^{-23}$ in the answering DCE's (GPC and GPA), self-synchronizing as in that Recommendation. The scrambler is reset at the start of phase 4 and after each renegotiation.

### 10.9 Quick start

A DCE may keep, for the last number dialled or answered, the loading and the symbol period of the last call, and offer them in INFO0 as the mode to start with. If its peer accepts in its INFO0, phase 2 is reduced to L1 of 0.5 s for a check of the loading, and the DCEs proceed to phase 3; this is in the manner of [ITU-T V.92]'s quick connect. A loading so offered shall still be checked by the receiver against the thresholds of clause 9.4.

### 10.10 Loading with amplitude: for further study

The MP message carries a loading value reserved for QAM constellations of 16 and 64 points per tone, with the 16-state four-dimensional trellis code of [ITU-T V.34] across tone pairs as [ITU-T G.992.1] applies it. Such a loading is for the transform coding modes of the EVS codec, which carry waveforms: through EVS at 48 kbit/s and above a tone's amplitude was measured to change by 0.3 to 1 dB rms from one symbol to the next, which would hold two levels 6 dB apart (Appendix II); through the ACELP codecs

it changes by 1.5 to 3 dB, and amplitude carries no bits. The loading's parameters are for further study. A DCE shall not request it unless both DCEs have indicated the capability in CM and JM, and shall fall back to DPSK when the peer does not confirm it in MP.

## 11 Error control and framing

### 11.1 Structure

The data of phase 4 pass, from the DTE towards the line, through:

1. [ITU-T V.42]: LAPM, with the V.42bis or V.44 compression the DCEs negotiate in XID (clause 12.1);
2. the inner code: the Reed-Solomon code of Annex C over the HDLC frames' octets, as a continuous octet stream (the LAPM frames with their flags are the stream; no further framing is added);
3. the interleaver of Annex C, which spreads each codeword over several modulation symbols so that an erased codec frame costs each codeword a fraction of its symbols;
4. the mapping of the coded bits onto the tone symbols of clause 11.4.

### 11.2 The code

The inner code shall be the Reed-Solomon code over GF($2^8$) of [ITU-T G.975] (the field of the primitive polynomial $x^8+x^4+x^3+x^2+1$, the code generator of its clause on the RS(255,239) code), shortened to the codeword lengths of Annex C: a codeword of $n$ octets with $k$ information octets and $n-k=2t$ check octets, correcting $t$ octets in error or $2t$ octets erased. The code rates of Table 4 are offered (Annex C shortens the codewords to 32 octets, so that the interleaver's delay stays short). A receiver should mark the octets of an erased codec frame as erasures (clause 11.3).

**Table 4** – Code rates of the inner code

| Rate | Codeword $n$ | Information $k$ | Corrects $t$ octets / $2t$ erasures |
|---|---|---|---|
| 7/8 | 32 | 28 | 2 / 4 |
| 3/4 | 32 | 24 | 4 / 8 |
| 1/2 | 32 | 16 | 8 / 16 |

### 11.3 Erased frames

A speech decoder conceals a lost or bad frame by extrapolating the previous frame's parameters: the output over the erased frame keeps the spectral envelope and the pitch of the signal before it and loses its information. A receiver of this Recommendation should detect an erased frame – from the phase errors of all its tones at once, or from the tones' levels decaying together as concealment attenuates – and mark the tone symbols of the symbol or symbols it spans as erasures to the Reed-Solomon decoder, which corrects twice as many erasures as errors. With the interleaving depth of Annex C one erased 20 ms frame erases one symbol in each of $D$ consecutive codewords' $D$ symbols, and is corrected at any code rate of Table 4; frame erasure ratios of the mobile networks are below 2 per cent ([3GPP TS 26.114]'s design targets), and the measured effect is in Appendix I.

### 11.4 Mapping

The interleaver's output bits shall be taken in order and mapped onto the loaded tones of a symbol from the lowest tone to the highest, two bits to a DQPSK tone and three to an 8-DPSK tone, Gray coded onto the phase difference as Table 5 shows. Bits left over at the end of a symbol are carried into the next. The receiver shall deliver, per tone symbol, the hard decision and an indication of its reliability (the distance of the received phase difference from the decision's), so that the Reed-Solomon decoder may be given erasures.

**Table 5** – Gray mapping of the phase difference

| **Phase difference** | 0 | 45 | 90 | 135 | 180 | 225 | 270 | 315 |
|---|---|---|---|---|---|---|---|---|
| DQPSK bits | 00 | – | 01 | – | 11 | – | 10 | – |
| 8-DPSK bits | 000 | 001 | 011 | 010 | 110 | 111 | 101 | 100 |

## 12 Link layer, compression and DTE interface

### 12.1 Link layer

Above the inner code the DCEs shall operate [ITU-T V.42] LAPM. The detection phase of [ITU-T V.42] (ODP and ADP) is not needed and shall not be used: both DCEs have indicated LAPM in the V.8 session; the originator shall start LAPM's XID exchange on entering phase 4. The DCEs should negotiate [ITU-T V.44] compression, or [ITU-T V.42bis] where V.44 is not available, in XID as those Recommendations specify. LAPM's T401 shall be set from the round-trip delay measured in phase 2 (clause 9.2), with at least 1 s added for the interleaver's delay and the symbol period; N401 should not exceed 64 octets at raw rates below 2400 bit/s, so that a frame fits in an interleaver block.

NOTE – The code of Annex C leaves LAPM an error ratio below $10^{-6}$ in the loaded modes; LAPM is retained for the residual errors, for the erased frames the code cannot cover, for flow control and for compression.

### 12.2 DTE interface

The DCE's DTE interface shall conform to [ITU-T V.250], with:

a) in `+MS` (modulation selection), the value `V_CVM` for this Recommendation's modulation, with its automode fallback to V.22 and V.21 (`+MS=V_CVM,1`); a DCE should make it its default on a speech path known to be a codec's. `+MS?` reports the value, `+MS=?` includes it in the list.
b) the extended command `+CVM` (Table 6) to constrain the mode set and to read the loading and the measurements of the current connection.
c) in the CONNECT result code, with `\V1` or `X` settings that give the rate, the coded data signalling rate (clause 5) and the text `/CVM`; the raw rate is read with `+CVM?`.
d) the result code `+CVMR: <raw>,<coded>` (unsolicited, if enabled by `+CVM`) on each renegotiation.

**Table 6** – The +CVM command

| **Command** | **Meaning** |
|---|---|
| `+CVM=<modes>[,<rate>[,<report>]]` | `<modes>`: the modes allowed, a sum of 1 (robust), 2 (narrowband), 4 (wideband), 8 (8-DPSK permitted), 16 (QAM loading permitted, clause 10.10); default all. `<rate>`: the signal interface rate, 8000 or 16000. `<report>`: 1 enables `+CVMR`. |
| `+CVM?` | `+CVM: <modes>,<rate>,<report>` and, while connected, a second line `+CVM: <raw tx>,<raw rx>,<coded tx>,<coded rx>,<symbol ms>,<tones tx>,<tones rx>,<code rate>,<renegotiations>,<erasures>`. |
| `+CVM=?` | the ranges. |

### 12.3 S-registers

S-register 44 (as a DCE of this Recommendation uses it) holds the number of tones loaded in the receive direction of the last call, S-register 45 the number of renegotiations of the last call. Both are read-only; the loading kept for the quick start of clause 10.9 is the DCE's own.

# 13 Interworking

## 13.1 With V series DCEs on the PSTN

A DCE of this Recommendation falls back to [ITU-T V.22] and [ITU-T V.21] (clause 8). Through a gateway between the mobile network and the PSTN the speech path has one codec (the mobile side's); V.22 passes AMR at 12.2 kbit/s with its data intact and V.21 passes every mode measured (Appendix I). A gateway that detects the V.8 session and switches its IP leg to [ITU-T G.711] or to the modem relay of [ITU-T V.150.1] changes nothing for this Recommendation: the codec that limits the path is the mobile side's.

## 13.2 With DCEs of this Recommendation across two codecs

A mobile-to-mobile call, or a call through a transcoding gateway, has two or more codecs in tandem in each direction. The probing of clause 9 measures the tandem as one channel; the loading follows the measurement. The throughput through two codecs is lower than through one (Appendix I gives the measured figures for AMR after AMR), and the retrain timers allow for the longer delay.

## 13.3 With facsimile

Facsimile terminals are served through Annex D: the DCE terminates the [ITU-T T.30] session on its side and carries it as [ITU-T T.38] packets in the data link. The modulations of a facsimile terminal (V.27ter, V.29, V.17) do not pass the codecs (Table I.2).

# 14 Performance

A DCE of this Recommendation shall meet the following, measured through the reference codecs of Appendix I with no frame erasures and the signal interface at 8000 Hz, with the method of that appendix:

a) through AMR at 12.2 kbit/s it shall connect in a narrowband mode with a raw rate of at least 1200 bit/s in each direction, and with the Reed-Solomon code at rate 3/4 or better;
b) through AMR at 7.95 kbit/s it shall connect in the robust mode (no mode of this Recommendation passes AMR at 4.75 kbit/s or AMR-WB at 6.6 kbit/s, whose radio conditions a network leaves only at the worst coverage: the DCE waits in the robust mode for a better codec mode);
c) through AMR-WB at 23.85 kbit/s with the interface at 16000 Hz it shall connect in a wideband mode with a raw rate of at least 2400 bit/s in each direction;
d) through EVS at 13.2 kbit/s narrowband it shall connect in the narrowband mode (the N400 comb) with a raw rate of at least 800 bit/s; through EVS at 24.4 kbit/s narrowband with a raw rate of at least 2000 bit/s; through EVS at 48 kbit/s wideband, with the interface at 16000 Hz, of at least 4000 bit/s;
e) the error ratio delivered to LAPM shall be below $10^{-6}$ at the loading chosen by clause 9.4;
f) a frame erasure ratio of 1 per cent (independent erasures) shall not lower the coded rate by more than the code rate step of Table 4.

The rates achievable are in Appendices I and II.

# Annex A

## Indication of this Recommendation's capabilities in V.8 and V.8bis

(This annex forms an integral part of this Recommendation.)

### A.1 Encoding in CM and JM

The CM and JM signals of [ITU-T V.8] are sequences of octets grouped in categories (call function, modulation modes, protocols, GSTN access, PCM modem availability, non-standard facilities). This Recommendation needs one bit of the modulation modes category to say "V.cvm available", and an information field of two octets to say what of it. The bit is an allocation that ITU-T makes in a revision of [ITU-T V.8]; until it is made:

a) the DCE shall send the information field of clause A.2 in the *non-standard facilities* category of CM and JM, as [ITU-T V.8] provides for it, with the country code and the provider code of [ITU-T T.35] that the implementer holds, followed by the octets of Table A.1 with the identifier octet `0x43` ('C') first, so that a DCE of this Recommendation recognizes the field among a provider's other uses of the category;
b) the DCE shall also set, in the modulation modes category, the bits of its fallback modulations (V.22 and V.21) and, in the protocols category, LAPM.

On the allocation of the bit, a DCE shall set it and shall still send the information field, in the category ITU-T assigns to it.

### A.2 The information field

**Table A.1** – The information field

| Octet | Bits | Meaning |
|---|---|---|
| 1 | 8–6 | version of this Recommendation: 001 |
| 1 | 5 | signal interface at 16000 Hz (1), or 8000 Hz (0) |
| 1 | 4 | facsimile of Annex D supported |
| 1 | 3 | QAM loading of clause 10.10 supported |
| 1 | 2–1 | reserved, 00 |
| 2 | 8 | robust mode (always 1) |
| 2 | 7 | narrowband mode (always 1) |
| 2 | 6 | wideband mode, 200 Hz comb |
| 2 | 5 | wideband mode, 400 Hz comb |
| 2 | 4 | 8-DPSK |
| 2 | 3 | 22.5 ms symbol (the 20 ms symbol is always supported) |
| 2 | 2 | quick start of clause 10.9 |
| 2 | 1 | reserved, 0 |

A DCE receiving an information field with a version it does not know shall treat the peer as having version 1 and the capabilities the field's bits it knows say.

### A.3 V.8bis

A DCE that uses the procedures of [ITU-T V.8bis] shall carry the same information field in the capabilities list of its MS and CL messages, in the non-standard capability of that Recommendation, with the same identifier octet.

# Annex B

## The probing signals and the INFO messages

(This annex forms an integral part of this Recommendation.)

### B.1 L1 and L2

L1 is each comb the transmitting DCE supports for the band of its signal interface (Table 3: N200, N300 and N400 at 8000 Hz; W200, W300 and W400 at 16000 Hz; S200, S300 and S400 at 32000 Hz), in the order of their codes, all tones loaded, DQPSK, with the 20 ms symbol, for 0.6 s each (30 symbols); L2 is the 200 Hz comb with the 22.5 ms symbol for 0.5 s (22 symbols). The phase difference transmitted on tone $k$ in symbol $n$ is given by two bits taken from the scrambler of clause 10 fed with ones, started at the first symbol of L1 and run across the tones from the lowest and across the symbols in order, so that the receiver, knowing the sequence, measures the phase error of every tone symbol. L1's first symbol is the phase reference. The level is −12 dBFS over all tones.

### B.2 What the receiver measures

For each tone $k$ and each of L1 and L2 the receiver computes the root mean square phase error $\sigma_k$ of the received phase differences against the transmitted ones, after removing the mean (the clock offset's rotation), the mean received level of the tone, and the fraction of tone symbols in error.

### B.3 INFO0

**Table B.1** – INFO0

| Octet | Bits | Meaning |
|---|---|---|
| 1 | 8–1 | the information field's octet 2 (Annex A): the modes supported |
| 2 | 8–1 | the combs this DCE will probe with, a bit each for codes 0 to 7 of Table 3 (bit 8 for code 0) |
| 3 | 8 | the same for code 8 |
| 3 | 7 | quick start offered (the loading of octets 5 onwards is the last call's) |
| 3 | 6 | quick start accepted (in the answering DCE's INFO0) |
| 3 | 5–1 | reserved, 0 |
| 4 | 8–1 | the DCE's transmit level in the data phases, in 0.5 dB below −9 dBFS |
| 5– | | with quick start: the loading of Table B.2 |

### B.4 INFO1 and MP

Each message is an HDLC frame of [ITU-T V.42] (flags, the FCS of ISO/IEC 13239, no address or control octet) carrying the octets above, transmitted twice in the robust mode; a receiver takes the first copy whose FCS checks. INFO0 is sent 100 ms after the end of the V.8 procedures by the calling DCE; INFO1 100 ms after the end of L2; the answering DCE's INFO0 50 ms after the end of the calling DCE's INFO1.

**Table B.2** – INFO1 and MP: the loading asked for

| Octet | Bits | Meaning |
|---|---|---|
| 1 | 8 | MP only: acknowledgement (the peer's MP was received) |
| 1 | 7–6 | symbol period: 00 20 ms, 01 22.5 ms |
| 1 | 5–4 | code rate: 00 1/2, 01 3/4, 10 7/8 |
| 1 | 3–1 | reserved, 0 |
| 2 | 8–1 | reserved, 0 |
| 3 | 8–1 | round-trip delay this DCE measured, in 10 ms, 255 if unknown |
| 4 | 8–5 | the comb: its code of Table 3 |
| 4 | 4–1 | reserved, 0 |
| 5 | 8–1 | number of tones *N* in the comb, 8 to 68 |
| 6– | | *N* two-bit fields, the lowest tone first, packed from bit 8: 00 not loaded, 01 DQPSK, 10 8-DPSK, 11 QAM (clause 10.10, with a following octet per tone for its constellation) |

# Annex C

## The inner code and its interleaver

(This annex forms an integral part of this Recommendation.)

### C.1 The code

The inner code is a Reed-Solomon code over GF($2^8$). The field is generated by the primitive polynomial $p(x) = x^8 + x^4 + x^3 + x^2 + 1$, and the code generator polynomial is

$$g(x) = \prod_{i=0}^{2t-1} (x - \alpha^i),$$

with $\alpha$ a root of $p(x)$: the field and the generator of the RS(255,239) code of [ITU-T G.975] (where $t = 8$), with $t$ as Table C.1 gives. Codewords are shortened to $n = 32$ octets: the encoder computes the check octets of the $k$ information octets preceded by $255 - n$ zero octets, which are not transmitted, and the decoder restores them. The first octet of a codeword is the highest order coefficient.

**Table C.1** – The code rates

| **Rate** | $n$ | $k$ | $t$ | **erasures corrected** ($2t$) |
|---|---|---|---|---|
| 7/8 | 32 | 28 | 2 | 4 |
| 3/4 | 32 | 24 | 4 | 8 |
| 1/2 | 32 | 16 | 8 | 16 |

### C.2 The interleaver

Codewords are interleaved in blocks of $D = 4$: the 4 codewords of a block are transmitted octet by octet across the codewords, that is, octet 0 of codeword 0, octet 0 of codeword 1, …, octet 0 of codeword 3, then octet 1 of each, and so on, 128 octets in all. Consecutive octets on the line thus belong to different codewords, and a symbol erasure of up to 4 octets (16 DQPSK tones) costs each codeword one octet. The interleaver's delay is one block: 128 octets, which is 0.64 s at 1600 bit/s and 0.31 s at 3300 bit/s. The receiver's de-interleaver is the inverse. The block boundary is marked by the first symbol of phase 4 and by the MP exchange of a renegotiation; the receiver recovers it, if lost, from the syndromes of the codewords (a block boundary error leaves every codeword uncorrectable).

### C.3 Erasure marking

The receiver should give the decoder, per octet, an erasure flag set when the tone symbols that carried the octet's bits were marked unreliable (clause 11.4) or lay in a frame detected as erased (clause 11.3). A decoder given $e$ erasures and $v$ errors corrects the codeword when $2v + e \leq 2t$.

# Annex D
## Facsimile

(This annex forms an integral part of this Recommendation.)

### D.1 Principle

A DCE that indicates facsimile in its information field (Annex A) terminates the [ITU-T T.30] session of the facsimile terminal on its side – as a Group 3 facsimile machine does on the PSTN, with the V.21, V.27ter, V.29 and V.17 modulations towards the terminal, or as a T.38 gateway does towards an IP network – and carries the session to its peer as the Internet facsimile protocol (IFP) packets of [ITU-T T.38] in the data link of this Recommendation. The peer DCE reproduces the session towards its own terminal. The procedure is that of a [ITU-T T.38] gateway pair with the data link of this Recommendation in place of the IP network, and of the facsimile relay of [ITU-T V.150.1] in its aim.

### D.2 Carriage of the IFP packets

Each IFP packet shall be sent in the information field of one LAPM I frame, preceded by one octet of type (`0x01`: T.38 IFP packet, other values reserved) and two octets of length (most significant first). The UDPTL transport of [ITU-T T.38] and its redundancy are not used: LAPM delivers the packets in order and without loss. The T.38 options in use shall be version 0, no error correction and no fill bit removal, unless both DCEs indicated otherwise in a T.38 options exchange carried the same way with type `0x02`.

### D.3 Rates

The slowest image modulation of [ITU-T T.30] is V.27ter at 2400 bit/s. Where the coded rate of the data link is below the modulation rate the terminals negotiate, the DCE shall accept only the error correction mode of [ITU-T T.30] Annex A from its terminal (it shall clear the non-ECM capability in the DIS it relays) and shall pace the sending terminal with the receive-not-ready procedure of that annex, as T.38 gateways do. The T.30 session's timers (T1, T2, T4) are met by the DCE towards its terminal; the DCE may spoof the terminal with preamble flags while the peer's answer is in transit, as [ITU-T T.38] allows.

# Appendix I

## The speech codecs as a channel: measurements

(This appendix does not form an integral part of this Recommendation.)

## I.1 Method

The measurements were made with the D-Modem project's tools [b-D-Modem], in October 2026, as follows.

a) **Codecs.** AMR: the OpenCORE AMR-NB library (encoder and decoder); AMR-WB: the VisualOn AMR-WB encoder and the OpenCORE AMR-WB decoder. Both are the implementations of the 3GPP reference code that Android carries; the AMR-WB pair was checked against the 3GPP reference code of [3GPP TS 26.173] on a comb signal (clause I.7). EVS: the 3GPP reference code of [3GPP TS 26.442] (fixed point, version 16.4.0), encoder and decoder, through files in the G.192 format. The codecs ran at the modes of Table I.1, without DTX unless stated, one encoding and decoding per direction.

b) **Signal interface.** 16 bit linear PCM at 8000 Hz (AMR, and AMR-WB through a 2:1 resampler with a 3400 Hz low pass, as a narrowband signal reaches a wideband codec), at 16000 Hz (AMR-WB and EVS as they are) or at 32000 Hz (EVS super wideband). The test signals were at −12 dBFS.

c) **Modulations of the V series.** Two D-Modem modems (`half_open8`) called each other through `netsim`, the project's network simulator, with the codec on its line (`NET_CODEC`), at each modulation of Table I.2 with the modems' default settings (V.42 and V.44 where the modulation carries them), and sent 10 s of data each way; a call passed when both modems reached CONNECT and the data arrived intact.

d) **Waveform probes.** Test signals were generated, passed through the codec (`codec_probe`, or the EVS programs) and analysed against the input, aligned by correlation: steady tones (the signal-to-noise ratio of the output against the input, with the best gain), binary FSK at 300 bit/s (bit errors of a non-coherent detector), DQPSK and 8-DPSK on one carrier (symbol errors and the root mean square phase error of the differential detector), and the combs of clause 10 with the phases changing at random every symbol (per tone: symbol errors, the rms phase error, and the rms change of the tone's amplitude from one symbol to the next). Each measurement is over 150 symbols or 800 bits; a tone counts as *good* when its symbol error ratio is below $2 \cdot 10^{-2}$, and the raw rate given is that of the good tones.

e) **Frame erasures.** Frames were erased at random, independently, at the ratio given, and the decoder told so (its concealment ran).

The scripts (`measure/probe.py`, `probe3.py`, `matrix.sh`) and the generator of the tables below (`tables.py`) are in the D-Modem repository with this document, so that the measurements can be repeated and extended.

**Table I.1** – The codec modes measured

| Codec | Modes (kbit/s) |
|---|---|
| AMR ([3GPP TS 26.090]), 8000 Hz | 4.75, 7.4, 7.95, 10.2, 12.2 |
| AMR-WB ([ITU-T G.722.2]), 16000 Hz | 6.6, 12.65, 15.85, 23.85; also behind a 2:1 resampler at 8000 Hz |
| EVS ([3GPP TS 26.445]), primary modes | NB (8000 Hz): 7.2, 9.6, 13.2, 24.4; WB (16000 Hz): 9.6, 13.2, 16.4, 24.4, 32, 48, 64, 96, 128; SWB (32000 Hz): 13.2, 24.4, 32, 48, 64, 96, 128 |
| EVS AMR-WB IO | 23.85 (16000 Hz) |

## I.2 The modulations of the V series through the codecs

Table I.2 gives, for each modulation and codec, the rate of the CONNECT the modems reached (bold where the data then arrived intact in both directions; with an [e] where the data had errors, which means the modulation carried no error correction, or that V.42's detection phase failed and the modems ran without it) and a dash where the modems did not connect. V.21 (and Bell 103) connect through every mode and

**Table I.2** – D-Modem's modulations through the codecs: CONNECT rate (bold: data intact; e: data in error; –: no connection)

| Codec | V.21 | Bell 103 | V.23 | V.22 | Bell 212A | V.22bis | V.32 | V.32bis | V.34 |
|---|---|---|---|---|---|---|---|---|---|
| none (G.711) | **300** | **300** | **1200** | **1200** | **1200** | **2400** | **9600** | **14400** | **33600** |
| AMR 12.2 | **300** | **300** | 1200[e] | **1200** | 1200[e] | 2400[e] | – | – | – |
| AMR 7.4 | 300[e] | 300[e] | 1200[e] | – | 300[e] | 2400[e] | – | – | – |
| AMR 4.75 | 300[e] | 300[e] | 1200[e] | – | 300[e] | – | – | – | – |
| AMR-WB 23.85 | **300** | 300[e] | 1200[e] | – | 300[e] | 2400[e] | – | – | – |
| AMR-WB 12.65 | 300[e] | 300[e] | 1200[e] | – | 300[e] | – | – | – | – |
| AMR-WB 6.6 | – | 300[e] | – | – | 300[e] | – | – | – | – |
| AMR 12.2, 1% of the frames erased | 300[e] | 300[e] | 1200[e] | 1200[e] | 300[e] | 2400[e] | – | – | – |
| AMR 12.2, 3% of the frames erased | 300[e] | 300[e] | 1200[e] | – | 300[e] | 2400[e] | – | – | – |
| AMR-WB 23.85, 1% of the frames erased | 300[e] | 300[e] | 1200[e] | – | 300[e] | – | – | – | – |

carry their 300 bit/s without error through AMR at 12.2 and AMR-WB at 23.85; at the lower modes their bits are in error (Table I.3). V.22 connects through AMR 12.2 and its LAPM comes up, so that its data arrives intact: 1200 bit/s. V.22bis connects through AMR 12.2 and AMR-WB 23.85 (its handshake is V.22's) but its 16-point constellation does not survive, nor does V.42's detection on it. V.32 and above do not complete their training; V.23's 1200 bit/s channel does, its 75 bit/s channel not.

**Table I.3** – Bit error ratio of 300 bit/s FSK through the codecs (8000 Hz interface)

| Codec | V.21 ch. 1 | V.21 ch. 2 | Bell 103 |
|---|---|---|---|
| none (G.711) | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ |
| AMR 12.2 | $< 10^{-4}$ | $< 10^{-4}$ | $< 10^{-4}$ |
| AMR 10.2 | 0.0139 | 0.0139 | 0.0076 |
| AMR 7.95 | $< 10^{-4}$ | 0.0038 | 0.0038 |
| AMR 7.4 | 0.0063 | 0.0038 | $< 10^{-4}$ |
| AMR-WB 23.85 | $< 10^{-4}$ | $< 10^{-4}$ | 0.0013 |
| AMR-WB 15.85 | $< 10^{-4}$ | 0.0013 | 0.0063 |
| AMR-WB 12.65 | 0.0013 | 0.0025 | 0.0089 |
| AMR 12.2 + AMR 12.2 | 0.0013 | $< 10^{-4}$ | $< 10^{-4}$ |
| AMR-WB 23.85 + AMR 12.2 | $< 10^{-4}$ | $< 10^{-4}$ | 0.0013 |
| AMR 12.2, 1% of the frames erased | 0.0228 | 0.0241 | 0.0304 |
| AMR 12.2, 3% of the frames erased | 0.0253 | 0.0253 | 0.0342 |

## I.3 Steady tones

Table I.4 gives the signal-to-noise ratio of a steady tone after the codec, which is high (25 to 37 dB through AMR 12.2) at most frequencies and low (10 to 18 dB) at a few: those whose period is a whole number of samples at 8000 Hz (800, 1000, 1333, 1600, 2000, 2667 Hz) and at 1500 Hz. A steady tone is the easiest signal for the long-term predictor once it has locked, and the hardest for the linear predictor, whose line spectrum it cannot place; the values are no guide to a modulated signal, whose phases the predictor must follow, and the single-carrier and comb measurements below are.

**Table I.4** – SNR (dB) of a steady tone at −12 dBFS after the codec, by frequency (Hz)

| Codec | 300 | 500 | 700 | 900 | 1100 | 1300 | 1500 | 1700 | 1900 | 2100 | 2300 | 2500 | 2700 | 2900 | 3100 | 3300 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| AMR 12.2 | 28 | 33 | 29 | 36 | 33 | 31 | 16 | 29 | 31 | 33 | 32 | 25 | 25 | 31 | 30 | 34 |
| AMR 7.4 | 28 | 17 | 26 | 30 | 26 | 27 | 15 | 24 | 30 | 22 | 15 | 16 | 33 | 18 | 18 | 19 |
| AMR 4.75 | 20 | 16 | 15 | 28 | 23 | 20 | 14 | 10 | 19 | 23 | 14 | 12 | 11 | 13 | 11 | 9 |
| AMR-WB 23.85 | 28 | 24 | 30 | 27 | 31 | 22 | 16 | 26 | 27 | 25 | 26 | 15 | 27 | 33 | 31 | 32 |
| AMR-WB 12.65 | 28 | 25 | 30 | 26 | 29 | 18 | 16 | 21 | 20 | 21 | 22 | 15 | 24 | 33 | 28 | 32 |
| AMR-WB 6.6 | 27 | 23 | 21 | 23 | 22 | 18 | 19 | 14 | 5 | 4 | 6 | 14 | 6 | 10 | 8 | 21 |
| AMR 12.2, 1% of the frames erased | 15 | 14 | 15 | -1 | 4 | 6 | 13 | 9 | 13 | 8 | 6 | 11 | 11 | 15 | 18 | 16 |
| AMR 12.2, 3% of the frames erased | 15 | 14 | 15 | -1 | 4 | 6 | 13 | 9 | 13 | 8 | 6 | 11 | 11 | 15 | 18 | 16 |
| AMR-WB 23.85, 1% of the frames erased | 14 | 14 | 12 | 13 | 12 | 14 | 11 | 14 | 13 | 13 | 14 | 11 | 14 | 13 | 13 | 9 |

## I.4 One carrier

Table I.5 gives the symbol error ratio and the rms phase error of DQPSK on one carrier at 1700 Hz, at 300 and 600 baud (600 baud is V.22's). Through AMR 12.2 and AMR-WB 23.85 the 600 baud signal has a symbol error ratio near $10^{-3}$ with 14° rms; through AMR 7.4 and below, and AMR-WB 12.65, it does not pass.

**Table I.5** – DQPSK on one carrier at 1700 Hz: symbol error ratio / rms phase error

| Codec | 300 baud (600 bit/s) | 600 baud (1200 bit/s) |
|---|---|---|
| none (G.711) | $< 10^{-3}$ / 2° | $< 10^{-3}$ / 3° |
| AMR 12.2 | $< 10^{-3}$ / 10° | 0.002 / 14° |
| AMR 10.2 | 0.010 / 15° | 0.036 / 20° |
| AMR 7.95 | 0.040 / 21° | 0.178 / 34° |
| AMR 7.4 | 0.064 / 24° | 0.192 / 36° |
| AMR-WB 23.85 | $< 10^{-3}$ / 10° | $< 10^{-3}$ / 12° |
| AMR-WB 15.85 | 0.002 / 13° | 0.033 / 19° |
| AMR-WB 12.65 | 0.007 / 15° | 0.047 / 23° |
| AMR 12.2 + AMR 12.2 | 0.004 / 13° | 0.017 / 20° |
| AMR-WB 23.85 + AMR 12.2 | 0.002 / 13° | 0.024 / 18° |
| AMR 12.2, 1% of the frames erased | 0.010 / 17° | 0.035 / 26° |
| AMR 12.2, 3% of the frames erased | 0.012 / 18° | 0.035 / 26° |

## I.5 The combs

Table I.6 gives, at the 8000 Hz interface, the tone symbol error ratio, the rms phase error and the raw rate of the good tones for the narrowband comb (200 Hz spacing, 400 to 3400 Hz, 16 tones) with the 20 ms and the 22.5 ms symbols, with 8-DPSK, with the shorter 12.5 ms symbol, and for the 400 Hz comb; Table I.7 the same at the 16000 Hz interface for the wideband combs (400 to 6800 Hz). The columns "+10 ms" are the 20 ms symbol with its boundaries 10 ms into the codec frame rather than at it. What the

**Table I.6** – Combs at the 8000 Hz interface: tone SER / rms phase error / raw rate of the good tones (bit/s)

| Codec | 200 Hz, 4-DPSK, 20 ms | 200 Hz, 4-DPSK, 20 ms, +10 ms | 200 Hz, 8-DPSK, 20 ms | 200 Hz, 4-DPSK, 22.5 ms | 200 Hz, 4-DPSK, 12.5 ms | 400 Hz, 4-DPSK, 20 ms |
|---|---|---|---|---|---|---|
| none (G.711) | $< 10^{-3}$ / 0° / 1600 | $< 10^{-3}$ / 0° / 1600 | $< 10^{-3}$ / 0° / 2400 | $< 10^{-3}$ / 0° / 1422 | $< 10^{-3}$ / 0° / 2560 | $< 10^{-3}$ / 0° / 800 |
| AMR 12.2 | 0.001 / 12° / 1600 | $< 10^{-3}$ / 12° / 1600 | 0.058 / 12° / 0 | $< 10^{-3}$ / 13° / 1422 | 0.054 / 23° / 160 | $< 10^{-3}$ / 7° / 800 |
| AMR 10.2 | 0.027 / 20° / 700 | 0.023 / 19° / 700 | 0.249 / 19° / 0 | 0.033 / 21° / 355 | 0.163 / 34° / 0 | 0.020 / 15° / 600 |
| AMR 7.95 | 0.146 / 32° / 0 | 0.139 / 32° / 0 | 0.457 / 33° / 0 | 0.172 / 35° / 0 | 0.368 / 54° / 0 | 0.055 / 23° / 200 |
| AMR 7.4 | 0.219 / 39° / 0 | 0.225 / 40° / 0 | 0.541 / 40° / 0 | 0.248 / 41° / 0 | 0.409 / 58° / 0 | 0.151 / 31° / 0 |
| AMR-WB 23.85 | $< 10^{-3}$ / 8° / 1600 | $< 10^{-3}$ / 8° / 1600 | 0.016 / 8° / 1650 | $< 10^{-3}$ / 7° / 1422 | 0.007 / 14° / 2080 | $< 10^{-3}$ / 6° / 800 |
| AMR-WB 15.85 | 0.011 / 14° / 1100 | 0.008 / 14° / 1400 | 0.129 / 14° / 750 | 0.007 / 13° / 1155 | 0.087 / 25° / 1120 | 0.001 / 11° / 800 |
| AMR-WB 12.65 | 0.060 / 21° / 700 | 0.059 / 21° / 700 | 0.265 / 21° / 300 | 0.048 / 20° / 622 | 0.169 / 35° / 320 | 0.017 / 17° / 500 |
| AMR 12.2 + AMR 12.2 | 0.033 / 20° / 500 | 0.034 / 20° / 400 | 0.256 / 21° / 0 | 0.037 / 21° / 266 | 0.163 / 34° / 0 | 0.001 / 12° / 800 |
| AMR-WB 23.85 + AMR 12.2 | 0.013 / 17° / 1200 | 0.018 / 17° / 900 | 0.178 / 17° / 0 | 0.016 / 17° / 888 | 0.127 / 31° / 0 | 0.002 / 11° / 800 |
| AMR 12.2, 1% of the frames erased | 0.026 / 20° / 300 | 0.035 / 24° / 0 | 0.094 / 23° / 0 | 0.022 / 20° / 444 | 0.070 / 28° / 0 | 0.030 / 21° / 0 |
| AMR 12.2, 3% of the frames erased | 0.044 / 26° / 0 | 0.057 / 29° / 0 | 0.117 / 27° / 0 | 0.047 / 27° / 0 | 0.070 / 28° / 0 | 0.048 / 26° / 0 |

combs show, and clause 10 builds on:

**Table I.7** – Combs at the 16000 Hz interface (AMR-WB): tone SER / rms phase error / raw rate of the good tones (bit/s)

| Codec | 200 Hz, 4-DPSK, 20 ms | 200 Hz, 8-DPSK, 20 ms | 400 Hz, 4-DPSK, 20 ms | 400 Hz, 8-DPSK, 20 ms | 300 Hz, 4-DPSK, 20 ms | 100 Hz, 4-DPSK, 20 ms |
|---|---|---|---|---|---|---|
| none (G.711) | $< 10^{-3}$ / 0° / 3400 | $< 10^{-3}$ / 0° / 5100 | $< 10^{-3}$ / 0° / 1700 | $< 10^{-3}$ / 0° / 2550 | $< 10^{-3}$ / 0° / 2300 | |
| AMR-WB 23.85 | 0.092 / 23° / 2800 | 0.198 / 26° / 900 | 0.087 / 20° / 1400 | 0.143 / 20° / 900 | 0.063 / 16° / 2000 | |
| AMR-WB 15.85 | 0.163 / 33° / 1000 | 0.397 / 36° / 300 | 0.129 / 30° / 600 | 0.319 / 29° / 300 | 0.083 / 22° / 1600 | |
| AMR-WB 12.65 | 0.279 / 46° / 600 | 0.502 / 47° / 0 | 0.233 / 40° / 300 | 0.476 / 41° / 150 | 0.148 / 31° / 700 | |
| AMR-WB 23.85, 1% of the frames erased | 0.111 / 29° / 1000 | 0.225 / 32° / 0 | 0.106 / 27° / 300 | 0.171 / 27° / 0 | 0.084 / 23° / 600 | |

- the 16-tone comb with DQPSK and the 20 ms symbol passes AMR 12.2 at a tone symbol error ratio of $10^{-3}$ (12° rms) and AMR-WB 23.85 better; it does not pass AMR 7.4 and below, nor AMR-WB 6.6;
- the position of the symbol in the codec frame makes no difference;
- 8-DPSK on the same comb has a symbol error ratio of $6 \cdot 10^{-2}$ through AMR 12.2: too high for the code; on the 400 Hz comb (8 tones) it passes, at 1050 bit/s, less than DQPSK on 16 tones;
- the shorter 12.5 ms symbol raises the error ratio by an order of magnitude: the codec needs the integration time;
- at the 16000 Hz interface through AMR-WB 23.85, 28 of the 34 tones are good and the tones above 6400 Hz are not: AMR-WB codes 6.4 to 7 kHz by bandwidth extension, without the waveform;
- the amplitude of a tone changes from one symbol to the next by 1.5 to 3 dB rms through the ACELP codecs: amplitude carries no bits there.

## I.6 Tandem codecs and frame erasures

Table I.6 also has AMR 12.2 followed by AMR 12.2 (a mobile-to-mobile call with a transcoding gateway, or two codecs in a call's path), and AMR-WB 23.85 followed by AMR 12.2 (a wideband terminal calling a narrowband one): the comb's error ratio roughly doubles, and the loading drops a few tones. With 1 and 3 per cent of the frames erased at random the comb's error ratio rises by about the erasure ratio itself (an erased frame is an erased symbol); the code of Annex C is dimensioned for this.

## I.7 Cross-check of the codec implementations

The AMR-WB encoder and decoder used (VisualOn, OpenCORE) were run on the wideband comb and compared with the 3GPP reference code of [3GPP TS 26.173] on the same input; the result is in the text of Table I.8.

**Table I.8** – The AMR-WB implementations against the 3GPP reference code

| Comparison | Result |
|---|---|
| OpenCORE AMR-NB encoder and decoder against the 3GPP TS 26.073 reference encoder and decoder, mode 12.2, 2 s of the N200 comb at 8000 Hz | decoded samples identical: 0 of 15 840 differ |
| VisualOn AMR-WB encoder and OpenCORE AMR-WB decoder against the 3GPP TS 26.173 reference coder and decoder, mode 23.85, 2 s of the W200 comb at 16000 Hz | decoded samples identical: 0 of 31 680 differ |

# Appendix II
## The EVS codec

(This appendix does not form an integral part of this Recommendation.)

## II.1 The codec

The enhanced voice services codec (EVS, [3GPP TS 26.445]) is the codec of VoLTE where both terminals have it, and the codec of 5G voice (VoNR, [b-GSMA NG.114]). It codes 20 ms frames at 5.9 to 128 kbit/s in four bandwidths (narrowband to 4 kHz, wideband to 8 kHz, super wideband to 16 kHz, full band to 20 kHz), choosing per frame between an ACELP core (speech), a transform core (MDCT: TCX and high-quality modes, for music and mixed content) and, at the low rates, a generic sound coding of the spectral envelope with a coarse residual; above the core's band it codes a bandwidth extension that carries no waveform. It also has an AMR-WB interoperable mode (the AMR-WB codec, bit-exact) and a channel-aware mode at 13.2 kbit/s with partial redundancy. Its signal classification decides which core codes a frame, so what it does to a modem's signal depends on what the signal looks like to it, not on the bit rate alone.

## II.2 Measurements

The combs of clause 10 were passed through the 3GPP reference encoder and decoder of [3GPP TS 26.442] (fixed point, version 16.4.0, through G.192 files, DTX off) at the signal interface rates and modes of Table I.1, as Appendix I describes. Tables II.1 to II.3 give, per comb, the tone symbol error ratio, the rms phase error, the raw rate of the good tones, and the rms change of a tone's amplitude from one symbol to the next in dB (the last for the question of amplitude modulation, clause 10.10).

**Table II.1** – EVS at the 8000 Hz interface (narrowband): tone SER / rms phase error / raw rate of the good tones (bit/s) / rms amplitude change (dB)

| Codec | 200 Hz, 4-DPSK, 20 ms | 200 Hz, 8-DPSK, 20 ms | 300 Hz, 4-DPSK, 20 ms | 300 Hz, 8-DPSK, 20 ms | 400 Hz, 4-DPSK, 20 ms | 400 Hz, 8-DPSK, 20 ms |
|---|---|---|---|---|---|---|
| EVS 7.2 | 0.457 / 70° / 300 / 8.8 | 0.664 / 73° / 0 / 9.1 | 0.504 / 74° / 300 / 8.0 | | 0.418 / 64° / 200 / 8.6 | 0.622 / 70° / 150 / 8.8 |
| EVS 13.2 | 0.036 / 18° / 1000 / 2.1 | 0.219 / 18° / 450 / 2.3 | 0.030 / 19° / 800 / 2.5 | 0.159 / 18° / 300 / 2.3 | $< 10^{-3}$ / 8° / 800 / 1.4 | 0.021 / 9° / 750 / 1.4 |
| EVS 24.4 | $< 10^{-3}$ / 6° / 1600 / 0.9 | 0.002 / 6° / 2400 / 1.0 | $< 10^{-3}$ / 4° / 1100 / 0.7 | $< 10^{-3}$ / 5° / 1650 / 0.7 | $< 10^{-3}$ / 5° / 800 / 0.8 | $< 10^{-3}$ / 5° / 1200 / 0.9 |
| EVS 9.6 | 0.259 / 45° / 100 / 5.4 | 0.527 / 46° / 0 / 5.2 | 0.178 / 36° / 0 / 4.7 | 0.448 / 34° / 0 / 4.4 | 0.138 / 32° / 200 / 4.4 | 0.339 / 32° / 0 / 4.1 |

**Table II.2** – EVS at the 16000 Hz interface (wideband); EVS AMR-WB IO is the codec's AMR-WB mode

| Codec | 200 Hz, 4-DPSK, 20 ms | 200 Hz, 8-DPSK, 20 ms | 300 Hz, 4-DPSK, 20 ms | 300 Hz, 8-DPSK, 20 ms | 400 Hz, 4-DPSK, 20 ms | 400 Hz, 8-DPSK, 20 ms |
|---|---|---|---|---|---|---|
| EVS 24.4 | 0.020 / 16° / 2200 / 2.9 | 0.177 / 17° / 600 / 2.7 | 0.001 / 10° / 2200 / 1.7 | 0.050 / 10° / 1350 / 1.7 | 0.001 / 9° / 1700 / 1.4 | 0.027 / 9° / 1650 / 1.4 |
| EVS 13.2 | 0.444 / 67° / 500 / 7.0 | 0.631 / 67° / 300 / 7.0 | 0.323 / 55° / 600 / 7.2 | 0.545 / 57° / 150 / 7.4 | 0.196 / 42° / 300 / 6.2 | 0.379 / 35° / 300 / 5.5 |
| EVS 48 | $< 10^{-3}$ / 6° / 3400 / 0.9 | 0.003 / 6° / 4950 / 1.0 | $< 10^{-3}$ / 5° / 2300 / 0.8 | 0.001 / 5° / 3450 / 0.7 | $< 10^{-3}$ / 3° / 1700 / 0.5 | $< 10^{-3}$ / 4° / 2550 / 0.5 |
| EVS 96 | $< 10^{-3}$ / 1° / 3400 / 0.3 | $< 10^{-3}$ / 1° / 5100 / 0.3 | $< 10^{-3}$ / 1° / 2300 / 0.3 | $< 10^{-3}$ / 1° / 3450 / 0.3 | $< 10^{-3}$ / 0° / 1700 / 0.3 | $< 10^{-3}$ / 1° / 2550 / 0.3 |
| EVS AMR-WB IO 23.85 | 0.118 / 27° / 2500 / 2.8 | 0.244 / 28° / 600 / 2.7 | 0.070 / 18° / 2000 / 2.0 | 0.113 / 17° / 1800 / 2.0 | 0.095 / 22° / 1400 / 2.2 | 0.178 / 22° / 600 / 2.4 |
| EVS 24.4, 1% of the frames erased | 0.041 / 24° / 400 / 3.4 | 0.201 / 24° / 0 / 3.3 | 0.021 / 19° / 900 / 2.9 | 0.075 / 18° / 450 / 2.4 | 0.022 / 21° / 700 / 2.0 | 0.055 / 19° / 150 / 2.1 |

**Table II.3** – EVS at the 32000 Hz interface (super wideband): the S200, S300 and S400 combs to 14 kHz

| Codec | 200 Hz, 4-DPSK, 20 ms | 200 Hz, 8-DPSK, 20 ms | 300 Hz, 4-DPSK, 20 ms | 300 Hz, 8-DPSK, 20 ms | 400 Hz, 4-DPSK, 20 ms | 400 Hz, 8-DPSK, 20 ms |
|---|---|---|---|---|---|---|
| EVS 24.4 | 0.390 / 61° / 1400 / 5.6 | 0.565 / 60° / 300 / 5.8 | 0.335 / 54° / 1500 / 5.0 | 0.473 / 54° / 900 / 5.0 | 0.348 / 55° / 1300 / 4.5 | 0.454 / 54° / 900 / 4.2 |
| EVS 48 | 0.124 / 28° / 3300 / 3.1 | 0.287 / 27° / 2250 / 3.1 | 0.087 / 22° / 3000 / 2.4 | 0.220 / 23° / 1950 / 2.6 | 0.079 / 18° / 3100 / 1.4 | 0.133 / 19° / 2250 / 1.4 |
| EVS 96 | $< 10^{-3}$ / 7° / 6900 / 1.0 | 0.008 / 7° / 8400 / 1.1 | $< 10^{-3}$ / 6° / 4600 / 0.9 | 0.003 / 6° / 6600 / 0.9 | $< 10^{-3}$ / 3° / 3500 / 0.6 | $< 10^{-3}$ / 3° / 5250 / 0.6 |
| EVS 128 | $< 10^{-3}$ / 3° / 6900 / 0.6 | $< 10^{-3}$ / 4° / 10350 / 0.6 | $< 10^{-3}$ / 3° / 4600 / 0.5 | $< 10^{-3}$ / 3° / 6900 / 0.5 | $< 10^{-3}$ / 2° / 3500 / 0.4 | $< 10^{-3}$ / 2° / 5250 / 0.4 |
| EVS 13.2 | 0.634 / 90° / 400 / 7.3 | 0.780 / 91° / 0 / 7.5 | 0.599 / 88° / 200 / 8.1 | 0.756 / 86° / 150 / 8.2 | 0.532 / 79° / 300 / 7.1 | 0.708 / 78° / 150 / 7.2 |

## II.3 What they show

a) **At 7.2 and 9.6 kbit/s** (narrowband) and **at 9.6 and 13.2 kbit/s wideband and super wideband**, the combs do not pass: the tone symbol error ratio is $10^{-1}$ or worse on every comb, and the amplitude of

a tone changes by 5 to 9 dB rms from one symbol to the next. The codec's low-rate generic coding keeps the spectral envelope and not the phases. The robust mode (300 bit/s FSK) passes: V.21's bit error ratio is below $10^{-2}$ at every mode but 9.6 wideband. A single carrier at 600 baud has a symbol error ratio of $3 \cdot 10^{-2}$ at 13.2 wideband: V.22 would connect and lose data.

b) **At 13.2 kbit/s narrowband**, the N400 comb passes with a tone symbol error ratio below $10^{-3}$ (8° rms, 800 bit/s raw), the N200 comb loads 10 of its 16 tones, and 8-DPSK does not hold. This is the mode a VoLTE call falls to in poor coverage; the probing's choice of the sparse comb is what keeps the DCE connected through it.

c) **At 24.4 kbit/s**, the transform core takes over for the comb and the picture changes: narrowband, every comb passes with 8-DPSK (N200: $2 \cdot 10^{-3}$, 6°, 2400 bit/s raw); wideband, W300 passes DQPSK at $10^{-3}$ (2200 bit/s) and W400 8-DPSK on 11 of 17 tones; super wideband, the band above about 8 kHz is the codec's bandwidth extension and carries nothing, and the probing loads 13 to 15 tones (1300 to 1500 bit/s) – less than the wideband interface gives at the same rate, because the signal's power is spread over tones the codec drops. A DCE at a 32000 Hz interface should therefore offer the wideband combs as well (Table 3: the probing finds the band).

d) **At 48 kbit/s and above**, the codec carries the comb nearly as a waveform: wideband, 8-DPSK on all 33 tones of W200 at $3 \cdot 10^{-3}$ (4950 bit/s raw) and on all tones of W400 without error; super wideband at 48 kbit/s, DQPSK on 31 of the 35 tones of S400 to 12.4 kHz (3100 bit/s); at 96 kbit/s, DQPSK on all 69 tones of S200 (6900 bit/s) and 8-DPSK on 56 of them (8400 bit/s); at 128 kbit/s, 8-DPSK on all 69 tones without error, 3.4° rms: 10 350 bit/s raw.

e) **Amplitude.** Through the ACELP codecs a tone's amplitude changes by 1.5 to 3 dB rms from one symbol to the next, and amplitude carries no bits. Through EVS at 24.4 kbit/s the change is 0.7 to 1.0 dB rms, at 48 it is 0.5 to 1.0 dB, at 96 and 128 it is 0.3 to 0.6 dB: two amplitude levels 6 dB apart, or the 16-point constellation of clause 10.10 with the trellis code, would hold at 48 kbit/s and above. With 16-QAM on the 69 tones of S200 the raw rate would be 13 800 bit/s, at 128 kbit/s of codec rate; the loading is for further study, as that clause says.

f) **AMR-WB IO** at 23.85 kbit/s behaves as the AMR-WB codec (which it is): W300 loads 20 of 23 tones (2000 bit/s), W200 25 of 33.

g) **Frame erasures** cost more through EVS than through AMR: with 1 per cent of the frames erased at 24.4 kbit/s wideband, the comb's tone symbol error ratio rises from $10^{-3}$ to $2 \cdot 10^{-2}$ (Table II.2), about twice the erasure ratio, because the transform core's overlap carries each erasure into the next frame as well: an erased frame costs two symbols. The receiver's erasure marking (clause 11.3) should mark both, and the probing should leave erased symbols out of its statistics, which are over a few tens of symbols.

h) **Mode changes.** A VoLTE call switches between 13.2 and 24.4 kbit/s with the radio conditions, and a VoNR call may hold higher modes. Between 13.2 and 24.4 the usable loading changes from one comb (N400, 800 bit/s) to another with 8-DPSK (N200, 2400 bit/s): the renegotiation of clause 9.6 is what follows the codec, and the DCE should expect it every few seconds in a moving terminal.

## II.4 Consequences for the signal set

The signal set of clause 10 needs no change for EVS: the combs, the two modulation orders and the probing's loading cover it from 13.2 kbit/s narrowband (N400, DQPSK) to 128 kbit/s super wideband (S200, 8-DPSK), a range of 800 to 10 350 bit/s raw. What EVS adds is the case for the super wideband combs (S200 to S400), which this Recommendation defines, and for the amplitude loading of clause 10.10, which it leaves for further study with the measured amplitude stability as its starting point. The terminal's part is in Appendix III: a terminal that can hold the EVS mode at 48 kbit/s or more for a DCE's call multiplies its throughput.

# Appendix III
## Terminals and networks

(This appendix does not form an integral part of this Recommendation.)

This appendix says what a terminal and a network should do so that a DCE of this Recommendation gets the most of the speech path; none of it is a requirement of this Recommendation on them.

### III.1 The terminal

a) **Signal interface.** The DCE needs the codec's PCM interface: in a UE, the point where the acoustic front end's output enters the encoder and the decoder's output leaves for the acoustic back end. The acoustic processing of [3GPP TS 26.131] (echo cancellation, noise suppression, automatic gain control, equalization) is for the acoustic path and should be bypassed for the DCE's samples; noise suppression in particular treats a comb as noise.
b) **Codec selection.** Where the terminal negotiates the codec (an IMS client, through SDP), it should offer, for a call the DCE will use, the highest modes available and the wideband codecs first: EVS at 24.4 kbit/s or more, AMR-WB at 23.85, AMR at 12.2; it should restrict the mode set (the `mode-set` parameter of [b-IETF RFC 4867]) to the highest modes, and set the EVS parameters `br` and `bw` of [3GPP TS 26.445] Annex A accordingly.
c) **DTX and VAD.** The terminal should turn discontinuous transmission off for the call (the `dtx` parameter of EVS; for AMR and AMR-WB it is the encoder's setting). A comb is not speech, and a voice activity detector may classify it as noise and send comfort noise in its place. The DCE's signals are continuous while a call is up, so that DTX, where it cannot be turned off, has nothing to detect as silence.
d) **Jitter buffer.** The jitter buffer management of [3GPP TS 26.114] adapts its delay by dropping or repeating frames, or by time-scaling the decoded signal ([3GPP TS 26.448] for EVS). Each adaptation is a step in the DCE's symbol timing of up to one frame, which the receiver re-acquires (clause 10.3); a terminal should prefer, for a DCE's call, a fixed delay large enough for the path's jitter.
e) **Level.** The terminal should pass the DCE's samples to the encoder at their level, without gain, and give the DCE the decoder's output at its level.

### III.2 The network

a) **Transcoding.** Each codec in the path costs throughput (Appendix I); transcoder-free operation between terminals with the same codec is best. A gateway to the PSTN transcodes to [ITU-T G.711], which costs nothing further.
b) **Echo cancellers.** Network echo cancellers of [b-ITU-T G.168] disable on the 2100 Hz tone with phase reversals that [ITU-T V.8]'s ANSam is; a DCE sends it as a matter of course. Mobile-to-mobile IMS calls have no network echo canceller; the UE's acoustic echo canceller is bypassed with the rest of the acoustic processing.
c) **Codec mode control.** The network changes the codec mode with the radio conditions; the DCE follows with renegotiations (clause 9.6). A network that holds a call at the highest mode while a DCE is in it gives it the highest throughput; the DCE cannot ask for that, and this Recommendation does not define a way.
d) **In-band signalling.** DTMF and other tones sent out of band ([b-IETF RFC 4733]) do not reach the speech path; the DCE's signals are not DTMF and are unaffected. A network that detects the V.8 tones and switches a PSTN gateway to a voice-band data mode ([b-ITU-T V.152]) or to modem relay ([b-ITU-T V.150.1]) does so on its own leg; the mobile leg's codec remains, and remains the limit.

# Appendix IV
## Implementation notes

(This appendix does not form an integral part of this Recommendation.)

A DCE of this Recommendation can be built from the parts of a V series software modem, as the D-Modem project [b-D-Modem] has them:

- the V.8 procedures, with the information field of Annex A added to the CM and JM encoders and decoders;
- the V.21 modulator and demodulator for the robust mode and the INFO and MP messages, and the HDLC framing of V.42 for their frames;
- a multitone modulator (an inverse discrete Fourier transform of the tone phases, the cyclic prefix prepended) and demodulator (the correlation of each tone over the useful part, as the probe scripts of Appendix I do it), with the differential detection, the Gray mapping of Table 5 and the scrambler of V.34;
- a Reed-Solomon encoder and decoder over GF($2^8$) for the code of Annex C, with erasure decoding; a block interleaver;
- the LAPM, V.42bis and V.44 of the V.42 implementation, unchanged, with the detection phase disabled;
- the V.250 command interpreter with the `+MS` value and the `+CVM` command of clause 12;
- for Annex D, the T.30 session machine and the T.38 packetizer of a T.38 gateway or an internet-aware fax implementation.

The computational load is small: the comb has at most 68 tones at 50 symbols/s, and the Reed-Solomon code at most 3300 bit/s. The probing's phase error estimates and the loading are the only parts not found in a V series modem, and they are a few hundred lines.

In D-Modem, the natural place is a data pump (`dsp/`) behind the existing pump interface, which the modem core (`core/`) already drives through V.8 for V.34 and V.90, with the link layer (`link/`) above it, and `netsim`'s `NET_CODEC` as the test line; the interoperability test harness then runs the DCE against itself through each codec mode as it runs the other modems today.

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[b-GSMA IR.92] GSMA PRD IR.92, *IMS Profile for Voice and SMS*.

[b-GSMA NG.114] GSMA PRD NG.114, *IMS Profile for Voice, Video and Messaging over 5GS*.

[b-D-Modem] D-Modem, a software modem (Neo-Desktop), `https://github.com/Neo-Desktop/d-modem-re`; the modems, the codec simulation (`dsp/test/codec_sim.cpp`, `netsim`) and the measurement scripts of Appendix I are in it.